B5-B6 / IMPLEMENTATION

Build and setup guide

Run the supplied Windows application

Extract the entire package to a writable local folder. Open B6 / Windows_x64 and double-click Run_SYNCLAB_Group_01.cmd. Keep StudentWorkshopApp.exe, the Qt/MinGW DLLs, platforms, sqldrivers and data folders together. This route includes its own Qt runtime.

Component	Delivered / verified
Platform	Windows x64
Application	C++17 / Qt 6.11.2 / MinGW 13.1.0
SQL runtime	Qt QSQLITE plugin
Launcher	Run_SYNCLAB_Group_01.cmd
Local data	Windows_x64/data/student_data.db
Initial contents	Schema v4; 10 workshops; no students or bookings

Build from the original source

• Install a Qt 6 Desktop kit with the matching compiler and QSQLITE driver.

• Open B5 / SYNCLAB / StudentWorkshopApp.pro in Qt Creator.

• Select the Qt 6 kit and a separate build folder. Build Release, then Run.

• Use the active .pro source list; older excluded split files are preserved but must not be added without reconciliation.

Both runnable and source routes are supplied. Startup is verified on this computer; confirm the normal display and database path on the assessment computer.

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Access, storage and packaging

First access

Students use Welcome without registration. A successful allocation returns a four-digit key for editing/deletion. Initial master and workshop passcodes are 0000 only when QSettings has no saved replacement. Change these defaults in system settings.

Bundled database location

The packaging manifest compiles SOURCE_DIR as a dot. The launcher changes the working directory to Windows_x64, so its supplied data/student_data.db is selected first. Keep that folder writable and launch through the supplied command file. Opening the executable from a different working folder can change path selection.

Original-source database order

Priority	Selected path
1	SOURCE_DIR/data/student_data.db, if present
2	Existing student_data.db in working folder
3	Existing student_data.db beside executable
4	Existing student_data.db in Qt local application data
5	Create SOURCE_DIR/data; use local-app fallback if needed

Settings and correspondence

QSettings stores passcodes, theme, wallpaper and notification addresses separately. Settings saved by a prior installation may still apply. Email actions open a mailto draft in a desktop email client; manual review/sending is required. A custom wallpaper is an external file.

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Verification and reproducible source

What changed for delivery

The original 73 application/source/resource files are preserved byte-for-byte in B5. B6 / Packaging_Source supplies a separate production manifest using the same C++ and resources, with packaging paths and SOURCE_DIR adjusted for the launcher. See its notes and packaging script for the exact transformation.

Executed delivery checks

The actual production executable started with PATH limited to Windows system folders. Required Qt, SQLite plugin and MinGW modules loaded from the shipped folder. The process remained running through the smoke check. The initial database passed integrity and foreign-key checks and contains only the ten workshop catalog rows.

Troubleshooting

• Extract the ZIP completely. Do not run a single executable inside the archive.

• Use the supplied launcher and a writable local folder. Restore a fresh test database only while the app is closed.

• For missing DLL/SQL-driver errors, restore the entire runnable folder.

• For source compilation, select Qt 6 and its matching compiler; excluded historical sources may have outdated APIs.

Evidence and limits

B6 contains deployment notes and smoke evidence. D9 contains compiled application tests and reproduction files. This is a portable folder bundle, with no installer. Normal GUI display, email handler and actual group demonstration need assessment-machine checks.

Implementation basis: StudentWorkshopApp.pro and its compiled sources. Prepared 07 October 2026. All demonstration identities and bookings are fictional.